Digital Filter

FIR	IIR
* $H(z) = h(0) + h(1) z^{-1} + h(2) z^{-2}$ OR $= a_0 + a_1 z^{-1} + a_2 z^{-2}$ 2nd order	$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$
- Found Relaization structure (Block diagram) with Direct method + Lattic	- There is Poles - Find Relizatian structure (Block diagram) with direct method only

⇒ For FIR :

* Diffrence equation from Transfere Function :

$$H(z) = \frac{y(z)}{x(z)} = h(0) + h(1) z^{-1} + h(2) z^{-2}$$

$$\therefore y(z) = h(0) x(z) + h(1) z^{-1} x(z) + h(2) x(z) z^{-2}$$

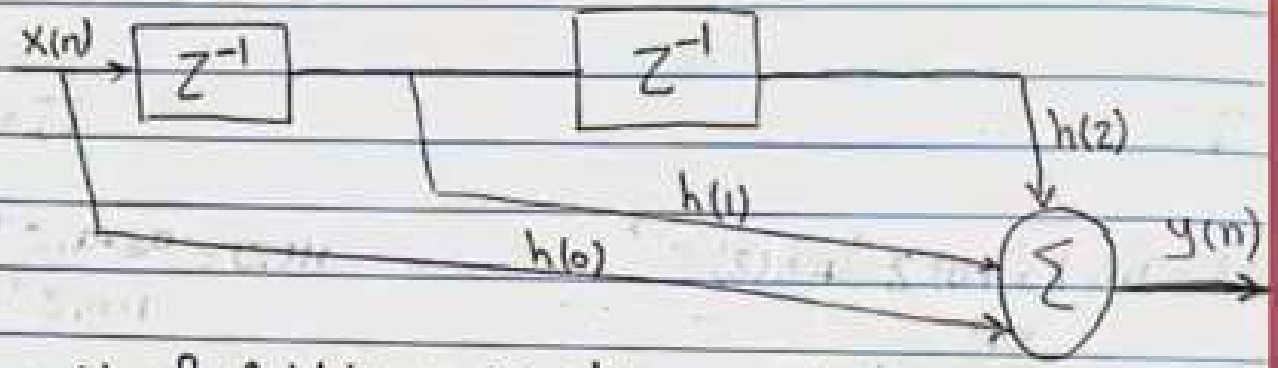
∴ Diffrence equation :

$$y(n) = h(0) x(n) + h(1) x(n-1) + h(2) x(n-2)$$

[2H]

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⇒ Block Diagram (Direct method) (Transversal)

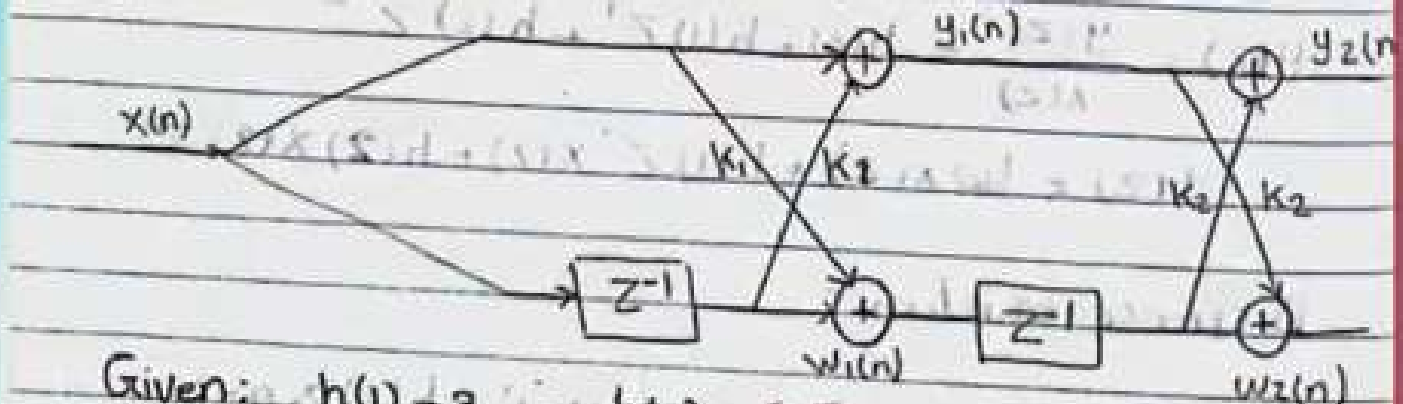


* No. of Addition = Input to Summer − 1
= 2

* No. of Mull. = 3

* Storage location = 2 + 2 + 3 = 7
(2 → I/P, 2 → Delay, 3 → Coeff (h(n)))

⇒ Lattice Methode



Given: h(1) = 3 h(2) = 8

Find K_1, K_2

* Difference equation for FLR:

$$y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2)$$

→ From lattice:

$$y_1(n) = x(n) + K_1 x(n-1) \quad \rightarrow ①$$

$$w_1(n) = K_1 x(n) + x(n-1) \quad \rightarrow ②$$

$$y_2(n) = y_1(n) + w_1(n-1) K_2 \quad \rightarrow ③$$

From ②: $w_1(n-1) = K_1 x(n-1) + x(n-2) \quad \rightarrow ④$

From (3,4) $y_2(n) = x(n) + K_1 x(n-1) + K_1 K_2 x(n-1) + K_2 x(n-$

$\therefore y_2(n) = x(n) + (K_1 + K_1 K_2) x(n-1) + K_2 x(n-2)$

Compare with Difference equation from transversal

$\therefore K_2 = h(2) = 5$

$h(1) = K_1 + (K_1 * K_2)$

$h(1) = K_1 + (K_1 \cdot h(2))$

$K_1 = \frac{h(1)}{1+K_2} = \frac{h(1)}{1+h(2)} = \frac{3}{1+5}$

$w(1) = K_1 (1+h(2))$

$K_1 = \frac{h(1)}{1+h(2)}$

OR Given K_1, K_2 → with the same method

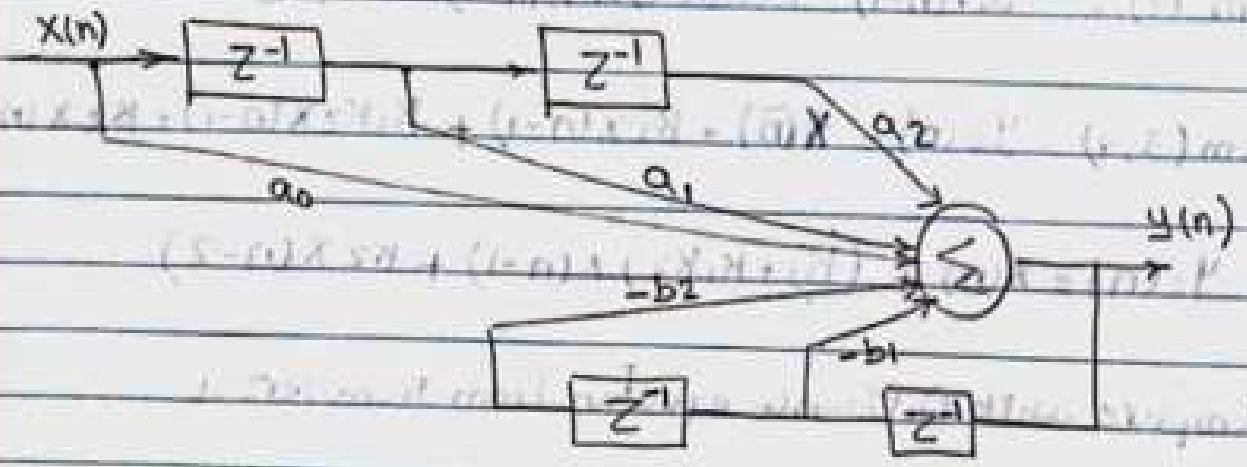
⇒ IIR Transversal method (Direct method):

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$\frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$Y(z)\left[1 + b_1 z^{-1} + b_2 z^{-2}\right] = X(z)\left[a_0 + a_1 z^{-1} + a_2 z^{-2}\right]$$

$$y(n) = a_0 x(n) + a_1 x(n-1) + a_2 x(n-2) - b_1 y(n-1) - b_2 y(n-2)$$

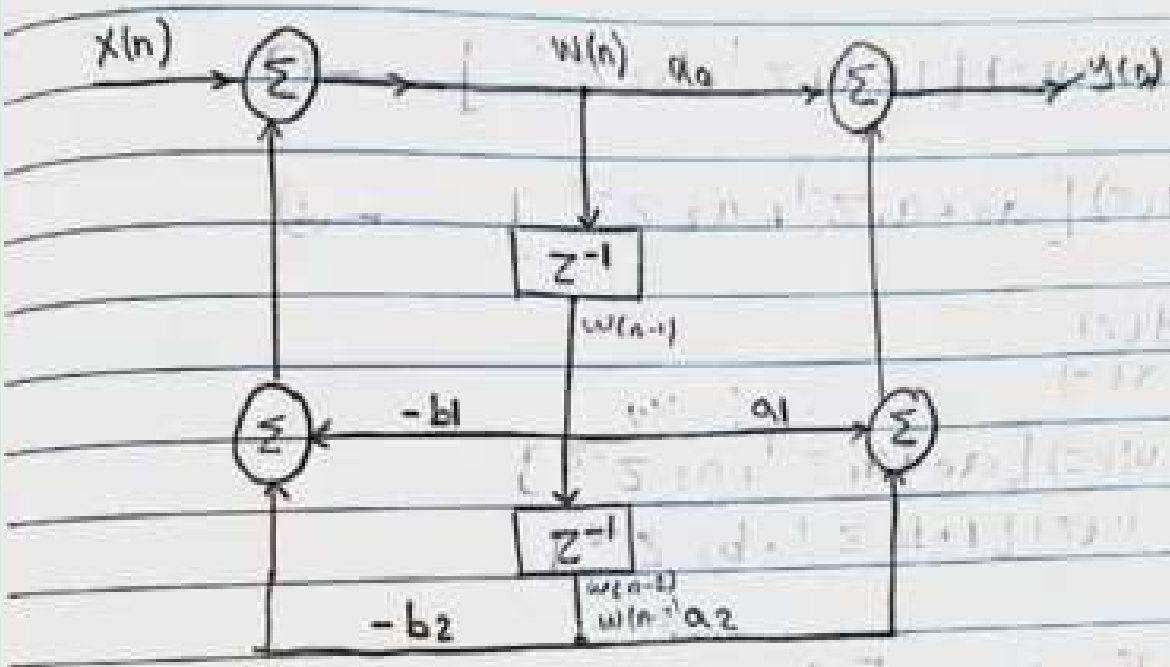


No. of Addition = 5 − 1 = 4

No. of mult. = 5

Storage = 2 + 4 + 5 = 11

⇒ Cascaded Filter for IIR:



[9] ⇒ Difference equation with two steps:

[1] Step [1] relation between x(n), w(n)

$$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$$

[2] Relation between w(n), y(n)

$$y(n) = a_0 w(n) + a_2 w(n-2) + a_1 w(n-1)$$

eq. 1, 2 called 2-step difference equation

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(b) Transfere function :

$$\Rightarrow W(z) = X(z) - b_1 w(z) z^{-1} - b_2 W(z) z^{-2}$$

$$\therefore X(z) = w(z) [1 + b_1 z^{-1} + b_2 z^{-2}] \longrightarrow (1)$$

$$\Rightarrow y(z) = W(z) [a_0 + a_1 z^{-1} + a_2 z^{-2}] \longrightarrow (2)$$

$$\therefore H(z) = \frac{y(z)}{X(z)}$$

$$H(z) = \frac{w(z) [a_0 + a_1 z^{-1} + a_2 z^{-2}]}{w(z) [1 + b_1 z^{-1} + b_2 z^{-2}]}$$

H(z) $\Rightarrow$ The Same T.f with Transversal method

Compare between FIR , IIR :

View of Comparison	FIR	IIR
• No. of Mult. = No. of Coefficient	12 h(0) ... h(11)	5 $a_0 \rightarrow a_2$ $b_1 \rightarrow b_2$
• No. of addition = No. of I/P, to Summetor - 1 	No. of I/P to Sumator = h(0) → h(11) = 12 No. of addition = 12-1 = 11	No. of I/P to Summetor = 2 No. of Summetor = 4 ∴ No. of addition = No. of Summetor = 4
Storage memory.	12 Coeff + 11 shift register + 1 O/P = 24	5 Coeff. + 2 Shift + 2 O/P = = 9

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Ex. Draw an determine the type of the following System:

$$H(z) = H_1(z) \times H_2(z)$$

لو واحد فلتر نوعه IIR يبقى ال System كله IIR ولو كلهم FIR يبقى نوعه FIR

Given: $H_1(z) = \sum_{n=0}^{11} h(n)\, z^{-n}$ FIR

$$H_2(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$ IIR

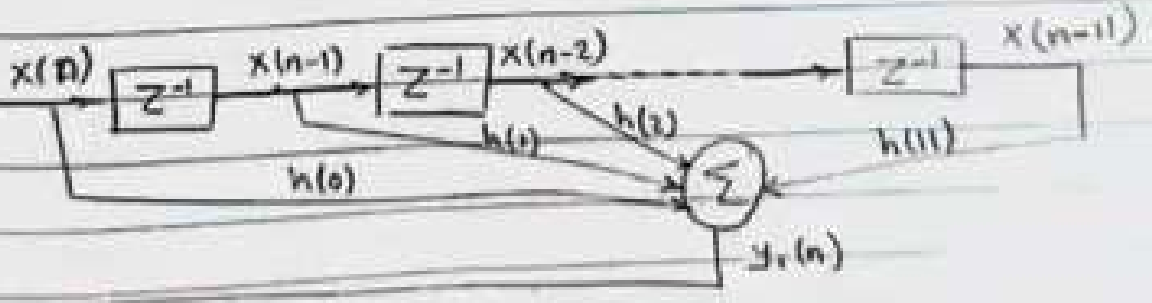
Sol.: 1 There is one IIR filter

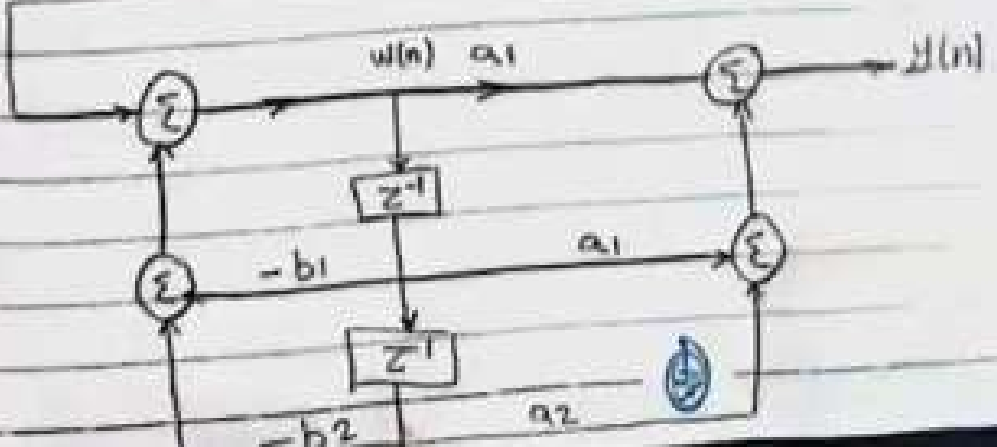
∴ System type is IIR.

2 To Draw Cascaded System



* o/p of first system is I/P for second system





To find diffrence eq.:

$$y_1(n) = x(n)h(0) + h(1)x(n-1) + h(2)x(n-2) + \dots + h(11)x(n-11)$$

$$w(n) = y_1(n) - b_1 w(n-1) - b_2 w(n-2)$$

$$y(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$$

] two stage diff. eq.

* No. of mult. = 17 ← No. of Coefficient

* No. of addition = 15 (مجموع)

* Storage memory = 31 (مجموع)

17 Coeff. 13 Shift Register 1 I/P

Ex.:

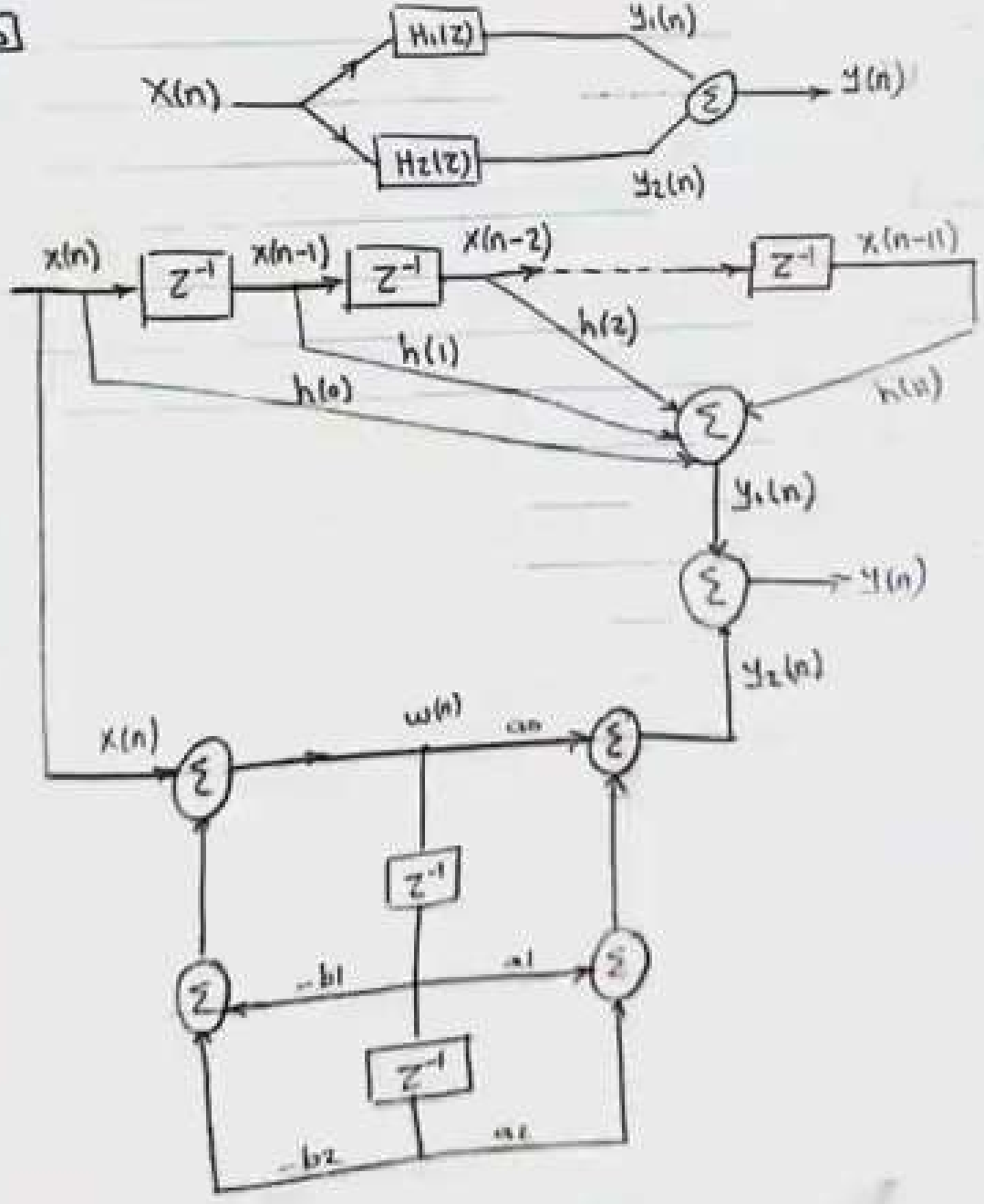
For $H(z) = H_1(z) + H_2(z)$ Find:

a) Type of system
b) Draw system realization (Block diagram)
c) Find difference eq.
d) Find No. of mult. + Addit

Sol

a) Type of system IIR

b)



[c] Diffrence eq.

$$y_1(n) = x(n)h(0) + h(1)x(n-1) + h(2)x(n-2) + \cdots + h(11)x(n-11)$$

$$y_2(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$$

$$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$$

$$y(n) = y_1(n) + y_2(n)$$

No. of mult. = 17

No. of add. = 16

Storage = 31

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